

2d. Non-exam assessment content – Iterative Design Challenge (J310/02, 03)

The non-exam assessment for this qualification requires learners to demonstrate their ability to draw together different areas of knowledge and understanding from across the full course of study through a single ‘challenge’ project – the ‘Iterative Design Challenge’.

At the heart of any iterative design challenge are three interrelated processes, requiring learners to:

- ‘explore’ needs
- ‘create’ solutions that demonstrate how the needs can be met
- ‘evaluate’ how well the needs have been met.

The above processes occur repeatedly as iterations throughout any process of designing prototyped solutions. This continual system of designing ensures constantly evolving iterations that build clearer needs and better solutions for a concept, so that the ideas and prototypes could be developed into successful products. Learners will be required as a result of their ‘Iterative Design Challenge’, to produce one final prototype or a set that work in conjunction.

Central to any iterative designing process is the thinking and management around the development of the processes (centre circle in Fig. 3). This requires learners to manage competing problems to prioritise requirements in order to progress with their challenge. Taking calculated risks and managing them

are an inherent part of this learning experience and should not be hidden, but encouraged as an integral and necessary aspect of iterative designing. Reflecting on problem solving through design iterations should be evidenced throughout.

The content of the non-exam assessment is laid out to clarify the iterative processes of ‘explore/create/evaluate’ and to highlight how the content requirements are not restricted to a single part/stage of an iterative designing challenge.

Learners are required to demonstrate their understanding that design and technology activity exists in contexts that influence the outcomes. Three different contextual challenges will be set by OCR each year in order for learners to explore a challenge that they can relate to. The challenges will:

- offer a broad range of real-world contexts, representing contemporary issues and concerns
- be open-ended, avoiding predetermining the materials or processes to be used to achieve a design solution
- focus on needs, wants and values of different groups, leading learners to address problems and/or opportunities
- be accessible and relevant to all learners regardless of areas they may have covered ‘in-depth’.

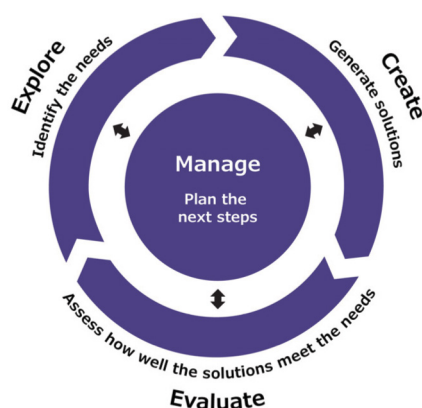


Fig. 3 Iterative Design Wheel showing key activities

Explore (AO1)

Exploring is about systematically understanding the need(s), known as requirements, of the stakeholder(s). The requirements should be described in a way that stimulates the 'create' stage of design development and forms the basis of measurable evaluation criteria in the 'evaluation' stage of the process. The requirements can be derived by exploring the following questions:

- *Who* are the stakeholders? e.g. using personas.
- *What* do stakeholders do and *when* do they do it? e.g. using task analysis.
- *Where* do stakeholders do it? e.g. through primary and secondary research that helps understand the physical, organisational, social and cultural environments.
- *Why* do stakeholders do what they do? e.g. establishing what stakeholders want to achieve.
- What is the impact of what stakeholders do on society (people), the environment (planet) and economics (profit)?